

Hands-on Lab: Create Tables using SQL Scripts and Load Data into Tables

Estimated time needed: 30 minutes

In this lab, you will learn how to run SQL scripts to create several tables at once, as well as how to load data into tables from .csv files.

Software Used in this Lab

In this lab, you will use [IBM Db2 Database](#). Db2 is a Relational Database Management System (RDBMS) from IBM, designed to store, analyze and retrieve the data efficiently.

To complete this lab you will utilize a Db2 database service on IBM Cloud. If you did not already complete this lab task earlier in this module, you will not yet have access to Db2 on IBM Cloud, and you will need to follow this lab first:

- [Hands-on Lab : Sign up for IBM Cloud, Create Db2 service instance and Get started with the Db2 console](#)

Database Used in this Lab

The database used in this lab is an internal database. You will be working on a sample HR database. This HR database schema consists of 5 tables called **EMPLOYEES**, **JOB_HISTORY**, **JOBS**, **DEPARTMENTS** and **LOCATIONS**. Each table has a few rows of sample data. The following diagram shows the tables for the HR database:

SAMPLE HR DATABASE TABLES

EMPLOYEES

EMP_ID	F_NAME	L_NAME	SSN	B_DATE	SEX	ADDRESS	JOB_ID	SALARY	MANAGER_ID	DEP_ID
E1001	John	Thomas	123456	1976-01-09	M	5631 Rice, OakPark,IL	100	100000	30001	2
E1002	Alice	James	123457	1972-07-31	F	980 Berry Ln, Elgin,IL	200	80000	30002	5
E1003	Steve	Wells	123458	1980-08-10	M	291 Springs, Gary,IL	300	50000	30002	5

JOB_HISTORY

EMPL_ID	START_DATE	JOBS_ID	DEPT_ID
E1001	2000-01-30	100	2
E1002	2010-08-16	200	5
E1003	2016-08-10	300	5

JOBS

JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY
100	Sr. Architect	60000	100000
200	Sr. Software Developer	60000	80000
300	Jr. Software Developer	40000	60000

DEPARTMENTS

DEPT_ID	DEPT_NAME	MANAGER_ID	LOC_ID
2	Architect Group	30001	L0001
5	Software Development	30002	L0002
7	Design Team	30003	L0003

LOCATIONS

LOC_ID	DEPT_ID
L0001	2
L0002	5
L0003	7

Objectives

After completing this lab, you will be able to:

- Create tables using SQL scripts
- Load data into tables

NOTE : Make sure that you are using the CSV file and datasets from the same instruction file.

Exercise 1: Create tables using SQL scripts

In this exercise, you will learn how to execute a script containing the CREATE TABLE commands for all the tables rather than create each table manually by typing the DDL commands in the SQL editor.

1. Download the script file to your computer:

- [HR_Database_Create_Tables_Script.sql](#)

2. Login to IBM Cloud and go to the [Resource List](#) where you can find the Db2 service instance that you created in a previous lab under **Services** section. Click on the Db2-xx service. Next, click on **Go to UI** button.

The screenshot shows the IBM Cloud console interface for a Db2-x4 service instance. The top navigation bar includes the IBM Cloud logo, a search bar, and links to Catalog, Docs, Support, and Manage. The main header shows the resource list with the instance name 'Db2-x4' and its status 'Active'. A sidebar on the left under the 'Manage' tab lists 'Getting started', 'Service credentials', and 'Connections'. The main content area features a 'Getting started' section with instructions on finding credentials and a 'Go to UI' button. To the right, a 'Need help?' section provides links to 'IBM answers' and 'Support'.

3. Click on **SQL** on the left corner and click the + icon

The screenshot displays the IBM Cloud SQL editor. On the left, a sidebar shows the 'SQL' tab selected, indicated by a red box and the number 1. The main editor area has a toolbar with a '+' icon for creating a new file, indicated by a red box and the number 2. The editor is titled '*Untitled - 1' and contains a single line of text.

Select the **From File** option.

Add new script

✕

Choose script source

Open a script to edit



Templates

Choose a template to start your SQL editor.



4. Locate the file **HR_Database_Create_Tables_Script.sql** that you downloaded to your computer earlier and open it.
5. Once the statements are in the SQL Editor tool , you can run the queries against the database by selecting the **Run All** button.

Run SQL

* HR_Databa... x

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```

37
38 CREATE TABLE JOBS (
39     JOB_IDENT CHAR(9) NOT NULL,
40     JOB_TITLE VARCHAR(30) ,
41     MIN_SALARY DECIMAL(10,2),
42     MAX_SALARY DECIMAL(10,2),
43     PRIMARY KEY (JOB_IDENT)
44 );
45
46 CREATE TABLE DEPARTMENTS (
47     DEPT_ID_DEP CHAR(9) NOT NULL,
48     DEP_NAME VARCHAR(15) ,
49     MANAGER_ID CHAR(9),
50     LOC_ID CHAR(9),
51     PRIMARY KEY (DEPT_ID_DEP)
52 );
53
54 CREATE TABLE LOCATIONS (
55     LOCT_ID CHAR(9) NOT NULL,
56     DEP_ID_LOC CHAR(9) NOT NULL,
57     PRIMARY KEY (LOCT_ID,DEP_ID_LOC)
58 );
59

```

Result - Feb 4, 2022 11:39:20 AM

- ✓ DROP TABLE JOBS
- ✓ DROP TABLE DEPARTMENTS
- ✓ DROP TABLE LOCATIONS
- ✓ -- Create the tables CREATE TABLE EMPLO...
- ✓ CREATE TABLE JOB_HISTORY (EMPL_ID ...
- ✓ CREATE TABLE JOBS (JOB_IDENT CHAR(...
- ✓ CREATE TABLE DEPARTMENTS (DEPT_ID_...
- ✓ CREATE TABLE LOCATIONS (LOCT_ID CH...

Run all ☐ Remember my selection

6. On the right side of the SQL editor window you will see a Result section. Clicking on a query in the Result section will show the execution details of the job like whether it ran successfully, or had any errors or warnings. Ensure your queries ran successfully and created all the tables.

- **Note:** You may see several errors before the successful creation of the tables. These errors relate to the dropping (removal) of any pre-existing version of these tables. You can ignore these errors.

7. Now you can look at the tables you created. Click on the data icon and then click on Tables tab

The screenshot shows a database management interface. On the left, a sidebar contains a menu with icons and labels: a hamburger menu icon, 'Dashboard', 'SQL Run SQL', 'Data' (highlighted with a red box), and 'Administration'. The main area has a top navigation bar with tabs: 'Load Data', 'Load History', 'Tables' (highlighted with a red box), 'Views', 'Indexes', 'Aliases', 'MQTs', 'Sequences', and 'Application objects'. Below the tabs is a search bar labeled 'Find schemas or tables'. Underneath the search bar is a section titled 'Schemas' containing a table with two columns: 'Name' and 'Type'. The table has one row with the value 'MYG36304' under 'Name' and 'User' under 'Type'.

Name	Type
MYG36304	User

8. Select the Schema corresponding to your Db2 userid. It typically starts with 3 letters (not SQL) followed by 5 numbers (but will be different from the **MYG36304** example below). Then on the right side of the screen you should see the 5 newly created tables listed RTMENTS, EMPLOYEES, JOBS, JOB_HISTORY and LOCATIONS (plus any other tables you may have created in previous labs e.g. PETSALE, PETRESCUE, etc.).

[Load Data](#)[Load History](#)**[Tables](#)**[Views](#)[Indexes](#)[Aliases](#)[MQTs](#)[Sequences](#)[Application objects](#)

Schemas

☒ Name	Type	Tables ▲
☒ MYG36304	User	5

Tables

☐ Name ▼	Schema
☐ DEPARTMENTS	MYG36304
☐ EMP	MYG36304
☐ JOBS	MYG36304
☐ JOBSHISTORY	MYG36304
☐ LOCATIONS	MYG36304

9. Click on any of the tables and you will see its Table Definition (that is, its list of columns, data types, etc).

Find schemas or tables

Schemas

Tables

New table +

☐ Name ▼	Schema	Properties
☐ DEPARTMENTS	MYG36304	...
☐ EMPLOYEES	MYG36304	...
☐ INSTRUCTOR	MYG36304	...
☐ JOBS	MYG36304	...
☐ JOBSHISTORY	MYG36304	...
☐ LOCATIONS	MYG36304	...

Total: 6, selected: 0

Table definition

EMPLOYEES

Name	Data type	Nullable
EMP_ID	CHAR	N
F_NAME	VARCHAR	N
L_NAME	VARCHAR	N
SSN	CHAR	Y
B_DATE	DATE	Y
SEX	CHAR	Y
ADDRESS	VARCHAR	Y

View data

Exercise 2: Load data into tables

In this exercise, you will learn how data can be loaded into Db2. You could manually insert each row into the table one by one, but that would take a long time. Instead, Db2 (and almost every other database) allows you to load data from .CSV files.

The steps below explain the process of loading data into the tables you created earlier in exercise 1.

1. Download the 5 .csv files below to your local computer:

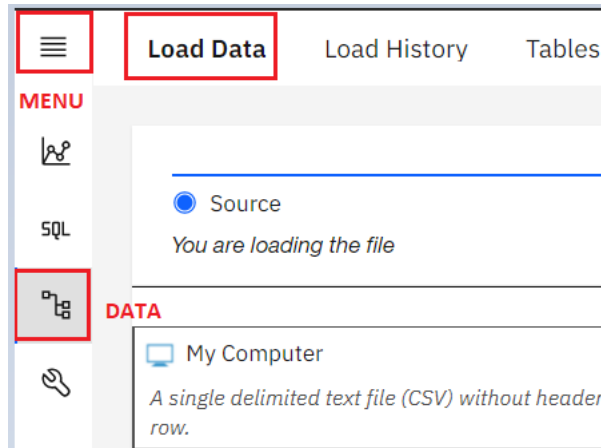
- [Departments.csv](#)
- [Employees.csv](#)
- [Jobs.csv](#)
- [Locations.csv](#)
- [JobsHistory.csv](#)

Note: For learners who are encountering issues with loading from .csv in Db2 using Firefox, they can download the .txt files and try with those. To download the .txt files, simply right-click on the file and select **Save link As** and save the file in local system.

- [Departments.txt](#)

- [Employees.txt](#)
- [Jobs.txt](#)
- [Locations.txt](#)
- [JobsHistory.txt](#)

2. In the Db2 Console, from the 3-bar menu icon in the top left corner, click **Load**, and then select **Load Data**.



3. On the **Load Data** page that opens, ensure **My Computer** is selected as the source. Click on the **browse files** link.

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Load Data

Load History

Tables

Views

Indexes

Aliases

MQTs

Sequences

Application objects

☒ Source

☐ Target

☐ Define

You are loading the file

My Computer

A single delimited text file (CSV) without header row.

Amazon S3

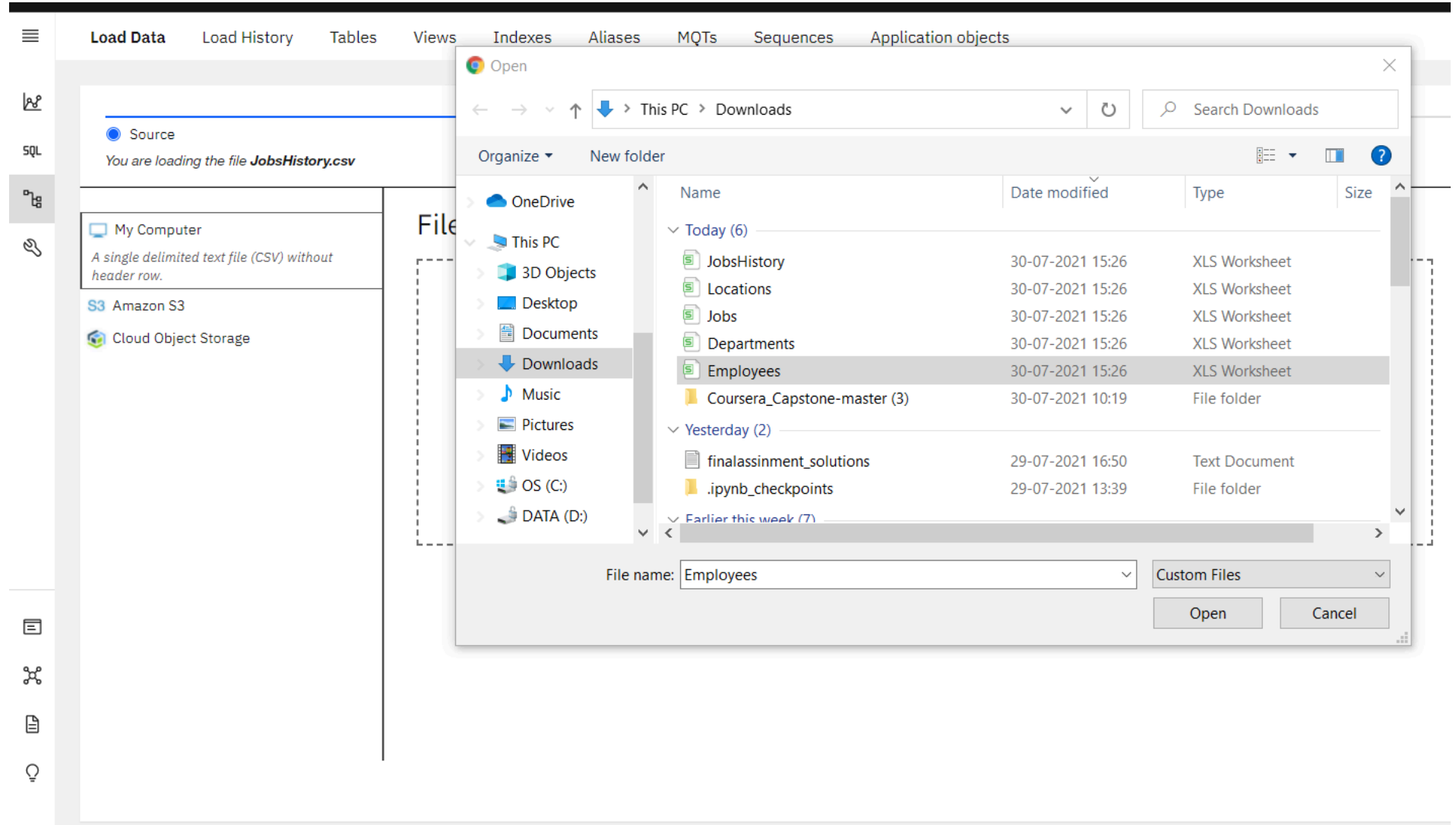
Cloud Object Storage

File selection



Drag a file here or [browse files](#)

4. Choose the file **Employees.csv** that you downloaded to your computer and click **Open**.



5. Once the File is selected, click **Next** in the bottom right corner.

Load Data Load History Tables Views Indexes Aliases MQTs Sequences Application objects

☒ Source ☐ Target ☐ Define ☐ File

You are loading the file **Employees.csv**

 My Computer
A single delimited text file (CSV) without header row.

 Amazon S3

 Cloud Object Storage

File selection



Drag a file here or [browse files](#)

6. Select the schema for your Db2 Userid (the one where you created the tables earlier). It will show all the tables that have been created in this schema previously, including the Employees table. Select the **EMPLOYEES** table, and in the new Table Definition tab that appears, choose **Overwrite table with new data** (note the warning message), then click **Next**. Select the **Employees** table.

Load Data Load History Tables Views Indexes Aliases MQTs Sequences Application objects

☒ Source

☒ Target

☐ Define

☐ F

You are loading the file **Employees.csv** into **HYL83142.EMPLOYEES**

Select a load target

Schema

Find schemas

HYL83142

✓

Table

New table +

Find tables in HYL83142

DEPARTMENTS

EMPLOYEES

JOBS

JOB_HISTORY

LOCATIONS

Table definition

EMPLOYEES

overwrite_option

☐ Append new data

☒ Overwrite table with

⚠

☐ All existing data will be removed. The load completes successfully.

COLUMN NAME
EMP_ID
F_NAME
L_NAME
SSN
B_DATE
SEX

7. Since the source data files do not contain any rows with column labels, **turn off** the setting for **Header in first row**.

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SQL

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Load Data

Load History

Tables

Views

Indexes

Aliases

MQTs

Sequences

Application objects

☑ Source

☑ Target

● Define

You are loading the file **Employees.csv** into **HYL83142.EMPLOYEES**

Code page (character encoding): 1208 (UTF-8)

⌵ ⓘ

 Separator: ,

Header in first row: ☐

 Time & date format: ⌵

Date format: YYYY-MM-DD

⌵ ⓘ

 Time format: HH:MM:SS

⌵ ⓘ

 Timestamp format: YYYY-MM-DD HH:MM:SS

⌵ ⓘ ⚠

	EMP_ID CHARACTER	F_NAME VARCHAR	L_NAME VARCHAR	SSN CHARACTER	B_DATE DATE	SEX CHARACTER	ADDRESS VARCHAR
1	E1001	John	Thomas	123456	01/09/1976	M	5631 Rice, OakPark,IL
2	E1002	Alice	James	123457	07/31/1972	F	980 Berry Ln, Elgin,IL
3	E1003	Steve	Wells	123458	08/10/1980	M	291 Springs, Gary,IL
4	E1004	Santosh	Kumar	123459	07/20/1985	M	511 Aurora Av, Aurora,IL
5	E1005	Ahmed	Hussain	123410	01/04/1981	M	216 Oak Tree, Geneva,IL
6	E1006	Nancy	Allen	123411	02/06/1978	F	111 Green Pl, Elgin,IL
7	E1007	Mary	Thomas	123412	05/05/1975	F	100 Rose Pl, Gary,IL
8	E1008	Bharath	Gupta	123413	05/06/1985	M	145 Berry Ln, Naperville,IL
9	E1009	Andrea	Jones	123414	07/09/1990	F	120 Fall Creek, Gary,IL
10	E1010	Ann	Jacob	123415	03/30/1982	F	111 Britany Springs,Elgin,IL

8. Click **Next**. Review the load settings and click **Begin Load** in the bottom right corner.

Load Data Load History Tables Views Indexes Aliases MQTs Sequences Application objects

☒ Source

☒ Target

☒ Define

☒ Fir

You are loading the file **Employees.csv** into **HYL83142.EMPLOYEES**

Review settings

Summary

Code page:	1208 (Default)
Separator:	, (Default)
Time format:	HH:MM:SS (Default)
Date format:	YYYY-MM-DD (Default)
Timestamp format:	YYYY-MM-DD HH:MM:SS (Default)
String delimiter:	(Default)

Option

Maximum number of warnings

1000

9. After loading has completed, you will notice that you were successful in loading all 10 rows of the Employees table. If there are any **Errors** or **Warnings**, you can see them on this screen.

Load details



My computer Target
Employees.csv HYL83142.EMPLOYEES

Status

Settings



10 **10** **0**
Rows read Rows loaded Rows rejected

Start time
07/30/2021 3:51:29 PM

End time
07/30/2021 3:51:34 PM

The data load job succeeded.
You can now work with your data.

Errors **0**

10. Click on the **Tables** tab and then select the **EMPLOYEES** table and then click on **View data**.

Load Data Load History **Tables** Views Indexes Aliases MQTs Sequences Application objects

Find schemas or tables

SQL

Schemas

Tables

New table +

Name	Schema	Properties
☐ DEPARTMENTS	HYL83142	...
☒ EMPLOYEES	HYL83142	...
☐ JOBS	HYL83142	...
☐ JOB_HISTORY	HYL83142	...
☐ LOCATIONS	HYL83142	...

Total: 5, selected: 1

Table definition

EMPLOYEES

Name	Data type	Nullabl
EMP_ID	CHAR	N
F_NAME	VARCHAR	N
L_NAME	VARCHAR	N
SSN	CHAR	Y
B_DATE	DATE	Y
SEX	CHAR	Y
ADDRESS	VARCHAR	Y
JOB_ID	CHAR	Y
SALARY	DECIMAL	Y
MANAGER_ID	CHAR	Y
DEP_ID	CHAR	N

View data

11. Now you can view the table data.



Load Data

Load History

Tables

Views

Indexes

Aliases

MQTs

Sequences

Application objects



HYL83142.EMPLOYEES

SQL



EMP_ID	F_NAME	L_NAME	SSN	B_DATE	SEX	ADDRESS	JOB_ID	SAL
E1001	John	Thomas	123456	1976-01-09	M	5631 Rice, OakPark,IL	100	1000
E1002	Alice	James	123457	1972-07-31	F	980 Berry Ln, Elgin,IL	200	8000
E1003	Steve	Wells	123458	1980-08-10	M	291 Springs, Gary,IL	300	5000
E1004	Santosh	Kumar	123459	1985-07-20	M	511 Aurora Av, Aurora,IL	400	6000
E1005	Ahmed	Hussain	123410	1981-01-04	M	216 Oak Tree, Geneva,IL	500	7000
E1006	Nancy	Allen	123411	1978-02-06	F	111 Green Pl, Elgin,IL	600	9000
E1007	Mary	Thomas	123412	1975-05-05	F	100 Rose Pl, Gary,IL	650	6500
E1008	Bharath	Gupta	123413	1985-05-06	M	145 Berry Ln, Naperville,IL	660	6500
E1009	Andrea	Jones	123414	1990-07-09	F	120 Fall Creek, Gary,IL	234	7000
E1010	Ann	Jacob	123415	1982-03-30	F	111 Britany Springs,Elgin,IL	220	7000

12. Now it's your turn to load data to the remaining 4 tables of the HR database **LOCATIONS**, **JOB_HISTORY**, **JOBS**, and **DEPARTMENTS** from the remaining source files.

13. Click **Load More Data** and then follow the steps from **Step 3** above again to load the remaining 4 tables.

IMPORTANT Make sure you perform the steps in **Step 7** for each of the 4 remaining file loads.

Congratulations! You have completed this lab, and you are ready for the next topic.

Author(s)

- [Rav Ahuja](#)
- [Sandip Saha Joy](#)



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